

Course Objectives

To provide practical experience in the operation and testing of synchronous and induction machines

Syllabus

List of Exercises /Experiments : (A minimum of **TWELVE** experiments are mandatory out of the fifteen listed.)

1. Load test on a three phase Slip Ring Induction Motor

Objectives:

- Start the motor using auto transformer or rotor resistance starter
- Plot the performance characteristics

2. No load and block rotor tests on a three phase Squirrel Cage Induction Motor

Objectives:

- Predetermination of performance parameters from circle diagram
- Determination of equivalent circuit

3. Starting of a three phase Squirrel Cage Induction Motor using Y-Starter Δ

Objectives:

- Start the motor using Y-Starter and perform load test Δ
- Plot the performance characteristics

4. Performance characteristics of a Pole Changing Induction Motor

Objectives:

- Run the motor in two different pole configurations (example 4 pole and 8 pole)
- Analyse the performance in the two cases by constructing circle diagrams and compare the results

~~5.~~ No Load and Blocked Rotor Tests on a single phase Induction Motor

Objectives:

- Conduct no load and blocked rotor tests on the motor
- Predetermine the equivalent circuit

6. Load Test on a single phase Induction Motor

Objectives:

- Perform load test on the motor
- Plot the performance characteristics of the motor

7. Variation of starting torque with rotor resistance in Slip-Ring Induction Motors

Objectives:

- Plot the variation of starting torque against rotor resistance in a three phase slip ring induction motor
- Find the external rotor resistance for which maximum starting torque is obtained.

8. V and inverted V curves of a Synchronous Motor

Objectives:

Plot the V and inverted V curves of the Synchronous Motor at no load and full load.

9. Regulation of a three phase Alternator by direct loading

Objectives:

- Determine the regulation of three phase alternator
- Plot the regulation versus load curve